# Exercise and Peer Review Instructions

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## Instructions for Exercises

- You may answer anonymously or include your name, as you prefer.
- Each assignment must be written independently. Collaborative discussion is encouraged, and using external resources—including web searches—is permitted, but all answers must be your own work and not copied.
- All work will be peer-reviewed (by yourself and randomly assigned peers). For expectations, grading rubric, and further details, read this document further. Summarising, a good answer should:
  - **Clearly explain the reasoning and process**, not just present final results.
  - **Be well-organised and concise**, with logical structure, precise technical language, and properly labelled code snippets/figures/tables as needed.
  - **Demonstrate factual correctness** with relevant, sufficient evidence and well-justified claims.
  - **Show critical analysis**, including justification of methods, discussion of alternatives or limitations, and integration of course concepts to the extend applicable.
  - **Reflect master's-level understanding**: not just procedural work, but explanation, synthesis, and professional communication.
  - **Not look like a code dump**: unless explicitly requested, you do not need to include program code in your answer.
- The language of submission and review is English.
- Upload a single, well-formatted PDF. Double-check before submitting that all figures and sections are present; Jupyter Notebooks can produce incomplete PDFs. We recommend R Markdown (or Quarto). For help, visit https://rmarkdown.rstudio.com/lesson-1.html and https://bookdown.org/yihui/rmarkdown/. It is a good idea to learn a system for writing proper reports, also for future studies and life after graduation. Incomplete or unreadable submissions may result in no points awarded.
- Answer the problems in the order given.
- Consult Moodle's general instructions and grading criteria before starting.
- Main textbook: ISLR_v2 ("An Introduction to Statistical Learning with Applications in R", 2nd edition, James et al.) or ISLP (Python version)—either is acceptable. You may use other materials as well.
- Submit as early as possible: you can revise and resubmit until the deadline. Due to peer review, we cannot accept late submissions. If you miss the deadline (and don't even submit empty PDF) you will loose points from the exercise set, including peer review points. Check in advance, well before the deadline date, that you can produce legible PDF files (especially with Jupyter notebooks students have often had last-minute surprises as PDF generation has now worked as expected). You have been warned.
- Work at your own pace, but refer to the study schedule on Moodle to avoid last-day rush.
- If you require special arrangements (e.g., sudden illness), inform staff promptly. By default, we assume you have followed the study schedule up to the time of notification; if you contact us at the last moment, we assume you have had time to do all exercises before the force majeure reason.
- If you find the problems difficult:
  - Follow the study schedule. Solving tasks after corresponding lectures reduces difficulty and stress.
  - Attend lectures and read the recommended textbook chapters.
  - Participate in exercise workshops and ask for help in workshops or the Team.
  - Talk with fellow students.

# Instructions for Peer Review

The course is split into three “blocks” of lectures and an Exercise Set.

The problems in the Exercise Set are meant to be solved during the corresponding block after the topics of the specific problems have been discussed in the lectures. At the end of each block, you should submit your answers for peer review, after which the model solutions and task-specific grading criteria are released. You can participate in a peer-review workshop during the following peer-review period, discussing the model answers and grading criteria. In the peer-review workshop, each student will review the solutions and answers of some randomly selected students and submit their reviews as instructed via Moodle.

Points will be awarded mainly based on peer reviews. The course staff reserves the right to override the peer review assessment. If you are not happy with the assessment you received, you can always ask the course staff to review your answers, after which the course staff can, if appropriate, override the peer reviews.

## Expectations for Master’s Level Work

As an EQF Level 7 (master’s level) course, DATA11002 expects students to demonstrate:

- **Highly specialised knowledge**: Applying machine learning concepts at a sophisticated level, showing awareness of current research and best practices
- **Critical awareness**: Evaluating methods critically, recognising limitations, and comparing alternative approaches
- **Specialised problem-solving**: Selecting and justifying appropriate methods for complex problems
- **Integration of knowledge**: Connecting concepts across different parts of the course and with other domains
- **Professional communication**: Articulating technical concepts clearly and precisely for peer audiences

Your answers should reflect this level of understanding. Simple application of formulas or procedures without critical reflection is insufficient for master’s level work.

## Generic Peer Review Criteria

If you are familiar with the Finnish school system, the grading criteria detailed below are similar to those in the Finnish high school matriculation examination for mathematics, natural sciences, and humanities (“reaaliaineet”).

The grading criteria below are meant to help you conduct peer reviews and inform you of what constitutes a good answer. Notice that individual exercises may contain specific instructions which will override these generic criteria. Additionally, the model answers may include task-specific grading guidance that takes precedence.

A good answer should **explain how the student obtained the results** rather than only reporting final solutions. This demonstrates that the student has understood the question, is familiar with the principles needed to solve the problem, and has followed appropriate procedures. The level of detail should be such that a fellow student can understand what has been done based on the written answer alone.

“Small” mistakes (typos, minor computation errors, etc.) should not significantly affect the assessment if it is evident that the student has understood the main principles and the question. However, substantial misunderstanding or incorrect methodology should be reflected in the assessment.

Assessment will be based on the written answer only. The assessment will be affected if the answer is missing necessary information, is ambiguous, or can be interpreted as wrong or incomplete by a reasonable person.

If the student uses figures, tables, or code snippets, it must be clear from the answer what conclusions the student draws from them. **About the role of code**: The solution should not look like a code dump! Unless specifically asked, the answer does not need to include the code used to produce the results. Small code snippets are acceptable if other parts of the solution make it clear what the student wants the reviewer to conclude from them. The reviewer is not required to search for results or missing details within code.

We strongly recommend that students use R Markdown, LaTeX, Quarto, or other typesetting software to produce the answer. However, if the solution contains hand-written parts, they should be clearly written: the reviewer is allowed to give zero points if they cannot easily read the answer due to unclear handwriting or poor scan quality.

The expression should be clear and concise. Recall that being able to communicate clearly about machine learning is one of the course's learning objectives!

## Three-Dimensional Rubric

Use the following rubric to guide your peer review. Consider each dimension separately, then form an overall assessment. The final assessment should be based on the answer as a whole, not a simple average of the three dimensions. Note that even one "poor" assessment typically indicates major issues that should result in a zero or a low overall score.

The model solutions will specify how many points can be awarded for each subtask. Use your judgement to determine the appropriate number of points based on these three dimensions and the overall quality of the answer.

### 1. Organisation and Clarity

This dimension evaluates how well the answer communicates ideas.

**Signs of a strong answer:**

- The answer is easily understandable for a fellow student at master's level
- Ideas are presented in a logical sequence that supports understanding
- The structure clearly relates to the question asked
- Writing is precise, clear, and uses appropriate technical terminology
- Mathematical notation follows standard conventions (when applicable)
- Figures, tables, and code snippets are properly labelled and referenced in the text
- The answer is appropriately concise—neither missing essential information nor including excessive irrelevant detail

**Signs of a weak answer:**

- Ideas are presented unclearly or inaccurately
- The answer can be interpreted as wrong or incomplete by a reasonable person
- Poor organisation makes it difficult to follow the reasoning
- Excessive verbosity obscures the main points
- Technical terms are used incorrectly or inconsistently

**Assessment levels:**

- **Poor**: It is unclear what the answer means even after reading it several times; major organisational problems impede understanding
- **Adequate**: Organisation is minimal but has some relation to the question; main points are identifiable but could be clearer
- **Excellent**: Writing is clear, logical, and well-organised around the question; technical communication is precise and professional

### 2. Evidence and Correctness

This dimension evaluates the factual content and reasoning.

**Signs of a strong answer:**

- The factual content is correct and relevant to the question
- All necessary information is included; the amount of detail is appropriate
- All claims made are substantiated with evidence or reasoning

- Source material (textbook, lectures) is used appropriately
- A clear distinction is made between facts, reasoned conclusions, and opinions
- Computational results are correct (or any errors are minor and do not reflect misunderstanding)
- When code is included, it is relevant and correctly implements the intended approach

**Signs of a weak answer:**

- The answer contains significant factual errors or misconceptions
- Essential information is missing
- Claims are unsubstantiated
- Computational or implementation errors suggest misunderstanding of core concepts

**Assessment levels:**

- **Poor**: The answer contains no relevant evidence or contains major factual errors that indicate misunderstanding
- **Adequate**: The answer includes some relevant evidence that addresses the question, though it may be incomplete or contain minor errors
- **Excellent**: There is correct, relevant evidence to support all aspects of the question; any computational work is accurate

### 3. Analysis and Critical Thinking

This dimension evaluates the depth of understanding and critical engagement with the material. It is particularly important at master's level.

**Signs of a strong answer:**

- The presented evidence and analysis directly address what the question asks
- The student demonstrates understanding of **why** certain methods or approaches are used, not just **how**
- Alternative approaches or limitations are considered when appropriate
- The student connects concepts to the broader context of machine learning
- For theoretical questions: derivations show understanding of underlying principles
- For applied questions: method choices are justified; results are interpreted critically
- The student demonstrates ability to synthesise knowledge from different parts of the course
- At master's level: evidence of critical thinking beyond mechanical application of formulas

**Signs of a weak answer:**

- The presented evidence or analysis is irrelevant to the question
- The answer shows only mechanical application without understanding
- The student has clearly misunderstood the problem
- No justification is provided for method choices or conclusions
- Results are reported without interpretation

**Assessment levels:**

- **Poor**: The evidence or analysis is not connected to the question; shows fundamental misunderstanding
- **Adequate**: The analysis addresses the question but may be superficial or lack critical engagement; shows basic understanding
- **Excellent**: The analysis clearly addresses all aspects of the question with appropriate depth; demonstrates critical thinking and synthesis appropriate for master's level

## Practical Tips for Conducting Peer Review

1. **Read the entire answer first** before making judgements—sometimes students explain their approach later in the text
2. **Compare with model solutions** but remember students may take different valid approaches
3. **Be constructive**: Note both strengths and areas for improvement in your feedback

4. **Be fair and consistent**: Apply the same standards to all answers you review
5. **Focus on understanding, not perfection**: Minor errors that don't reflect misunderstanding should not heavily penalise an otherwise good answer
6. **Consider the learning objectives**: Does the answer demonstrate the intended learning?

Remember that peer review is a learning opportunity for both the reviewer and the reviewed. Thoughtful, constructive feedback helps everyone improve.

## Handling Problematic Cases

**What if the answer is completely off-topic?** Assess as "Poor" in all dimensions and award minimal points. Provide brief feedback noting the answer does not address the question.

**What if I can't understand the handwriting or the pdf is corrupted?** You may assess the answer negatively for Organisation and Clarity; you are allowed to give zero points. Note in your feedback that the submission was illegible.

**What if the answer is partly correct, how many points should I give?** Good answers (full points) and empty or really bad answers (zero points) are usually easy to review. There is often no single objective rule for giving points in the middle cases: please just use your judgement and give points that best describe the student's knowledge. Use your judgement to determine which level best describes each dimension. The overall assessment should reflect the proportion of the question that was answered correctly.

**What if the student used a different method than the model solution?** This is fine if the alternative method is valid and correctly applied. Assess based on correctness of the chosen approach and quality of explanation.

**What if I'm unsure about the correctness?** Focus your assessment on the clarity of explanation and logical reasoning. If the approach seems sound but you're uncertain about technical details, note this in your feedback.

**What if the answer looks like a code dump and I can't understand the answer without reviewing the code?** You do not need to fish for the answers or missing details from the program code. In such cases, you are allowed to give zero points. Provide brief feedback noting that undrestanding the answer would have required debugging the source code.

**What if I don't want to make a call?** Always remember that the student can ask the course staff to review the grading. Therefore, the course staff (not you) always has the final responsibility.

**What if you suspect foul play?** If you suspect unethical or forbidden behaviour (e.g., forbidden use of generative AI, copying of answers, etc.) please email the course staff.

**What if I need help in reviewing?** You can ask for help in reviewing in the exercise workshop or in the course Team.